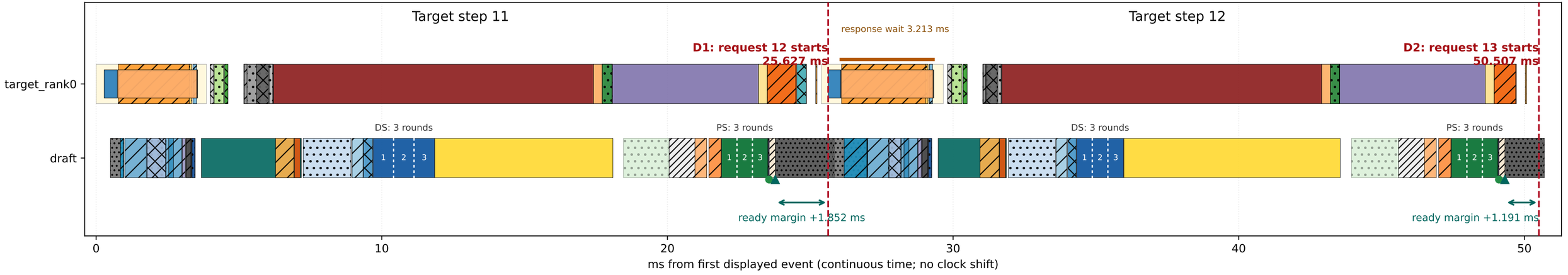
LayerSkip Llama2-7B + llama-68m | Exit 21; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | P1 hit -> P1 hit | UID 461_t0 | steps 11-12

Fresh local orion RTX4090; B1 BF16 T0.7; 12 fixed turns x 2; first10/terminal/cap excluded; GPU forward markers enabled | Exit 21; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | B=1, K1/K2 3/3

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

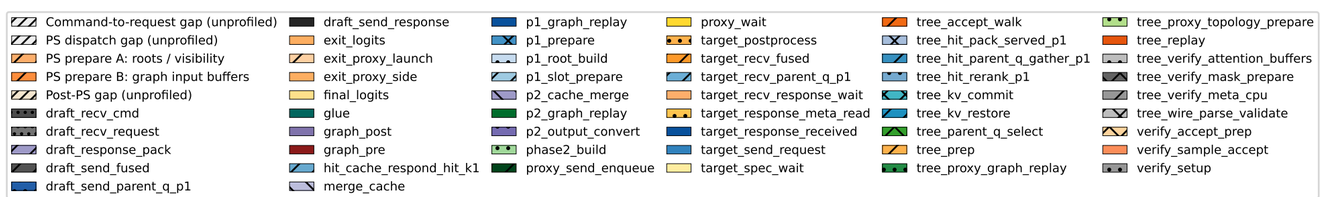
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	698	42.329	2.292	88.405
P2 hit	712	43.178	2.409	93.290
Miss	239	14.494	1.904	75.899
Overall	**1649**	**85.506**	**2.286**	**88.754**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	698	25.929	0.191	7.580	+7.008	0.168	2.153	+1.648
P2 hit	712	25.819	0.200	6.266	+5.666	0.168	2.488	+1.983
Miss	239	25.083	0.169	7.864	+7.359	0.169	2.478	+1.972
Overall	**1649**	**25.759**	**0.192**	**7.054**	**+6.479**	**0.168**	**2.345**	**+1.840**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

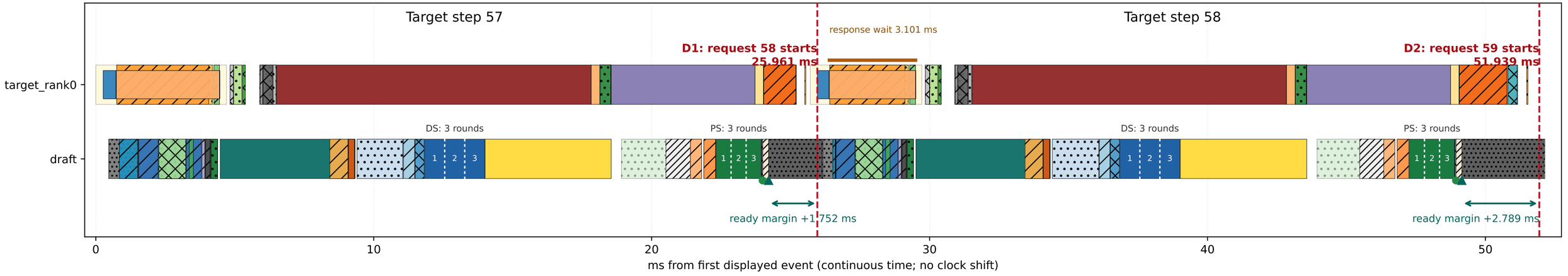
LayerSkip Llama2-7B + llama-68m | Exit 21; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | P2 hit -> P2 hit | UID 341_t0 | steps 57-58

Fresh local orion RTX4090; B1 BF16 T0.7; 12 fixed turns x 2; first10/terminal/cap excluded; GPU forward markers enabled | Exit 21; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | B=1, K1/K2 3/3

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

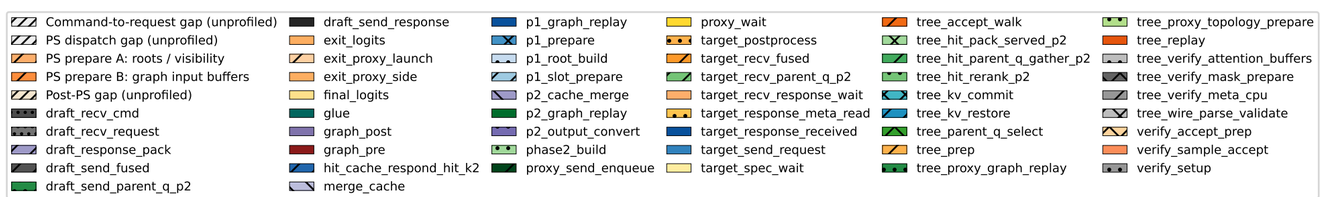
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	698	42.329	2.292	88.405
P2 hit	712	43.178	2.409	93.290
Miss	239	14.494	1.904	75.899
Overall	**1649**	**85.506**	**2.286**	**88.754**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	698	25.929	0.191	7.580	+7.008	0.168	2.153	+1.648
P2 hit	712	25.819	0.200	6.266	+5.666	0.168	2.488	+1.983
Miss	239	25.083	0.169	7.864	+7.359	0.169	2.478	+1.972
Overall	**1649**	**25.759**	**0.192**	**7.054**	**+6.479**	**0.168**	**2.345**	**+1.840**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

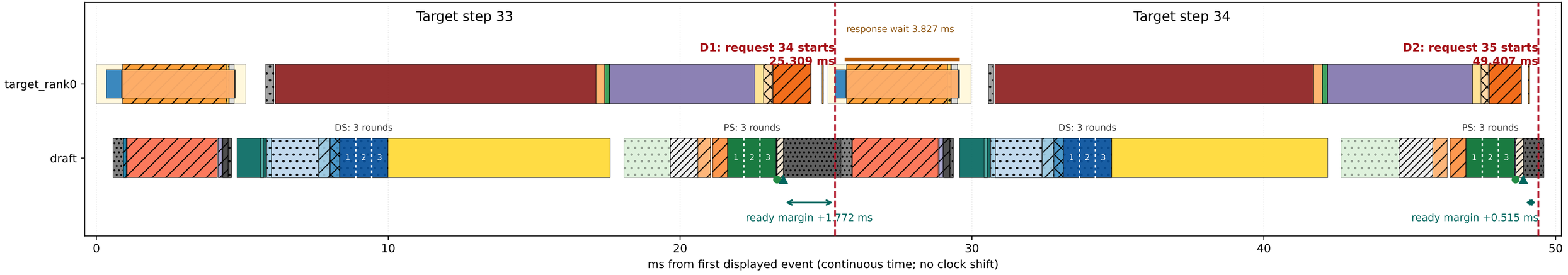
LayerSkip Llama2-7B + llama-68m | Exit 21; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | Miss -> Miss | UID 541_t0 | steps 33-34

Fresh local orion RTX4090; B1 BF16 T0.7; 12 fixed turns x 2; first10/terminal/cap excluded; GPU forward markers enabled | Exit 21; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | B=1, K1/K2 3/3

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

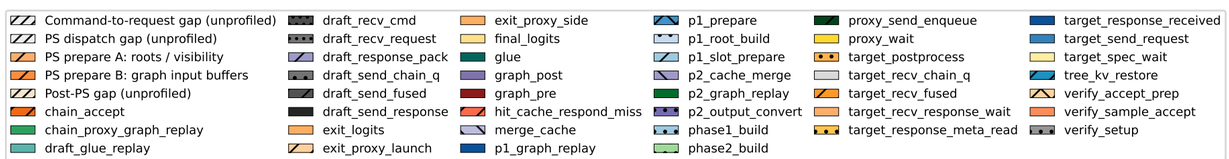
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	698	42.329	2.292	88.405
P2 hit	712	43.178	2.409	93.290
Miss	239	14.494	1.904	75.899
Overall	**1649**	**85.506**	**2.286**	**88.754**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	698	25.929	0.191	7.580	+7.008	0.168	2.153	+1.648
P2 hit	712	25.819	0.200	6.266	+5.666	0.168	2.488	+1.983
Miss	239	25.083	0.169	7.864	+7.359	0.169	2.478	+1.972
Overall	**1649**	**25.759**	**0.192**	**7.054**	**+6.479**	**0.168**	**2.345**	**+1.840**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.